MILITARY SPECIFICATION

# TRANSMITTER, RADIO
# T-1458/DKT-59

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification establishes the performance, design, development and test requirements for the T-1458/DKT-59, Radio Transmitter critical item referred to herein as the transmitter.

1.2 Classification. The transmitters covered by this specification are classified as specified (see 6.2.1):

Type I - 225 Megahertz (MHz) to 260 MHz

Type II - 1435 MHz to 1540 MHz

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, New Jersey 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-E-5400 | Electronic Equipment, Airborne, General Specification for |
| MIL-T-5422 | Testing, Environmental, Airborne Electronic and Associated Equipment |
| MIL-E-17555 | Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of |
| MIL-T-18303 | Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment; Format for |
| MIL-N-18307 | Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment |
| MIL-T-85650 | Transmitting Set, Telemetric Data AN/DKT-59 |
| MIL-A-85652 | Amplifier-Mixer AM-7157/DKT-59 |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of U.S. Military Property |



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| | |
|---|---|
| MIL-STD-461 | Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement of |
| MIL-STD-704 | Aircraft Electric Power Characteristics |
| MIL-STD-781 | Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution |
| MIL-STD-794 | Parts and Equipment, Procedures for Packaging and Packing of |
| MIL-STD-831 | Test Reports, Preparation of |
| MIL-STD-883 | Test Methods and Procedures for Microelectronics |
| MIL-STD-965 | Parts Control Program |
| MIL-STD-2078 | Requirements for Preparation of Support Equipment Depot Level Rework Standards |
| MIL-STD-45662 | Calibration System Requirements |

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

| | |
|---|---|
| 1356AS313 | Transmitter, Radio |



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| | |
|---|---|
| INTER RANGE INSTRUMENTATION GROUP (IRIG) STANDARD 106-80 | Telemetry Standards |

(Application for copies should be addressed to the Department of Defense Technical Information Center, Cameron Station, Alexandria, VA 22314.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Transmitter description. The transmitter shall be a solid state, crystal-stabilized, true frequency modulation (FM) telemetry transmitter and shall be capable of transmitting the intelligence contained in the frequency division multiplex composite signal supplied by the amplifier-mixer, MIL-A-85652.

3.1.1 Interface definition. The transmitter's physical and electrical interfaces with the module assembly unit shall be in accordance with Drawing 1356AS313.

3.2 Characteristics.

3.2.1 Performance. The transmitter shall meet the performance requirements specified herein under the conditions of 3.2.5.

3.2.1.1 Operational.

3.2.1.1.1 External power. The transmitter shall operate using external power with the characteristics as specified in MIL-STD-704 and modified herein as follows:

a. 24.0 to 32.0 volts direct current (Vdc)

b. Transient surge to +40 to -40 Vdc



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c. Maximum current shall be not greater than 2.0 amperes for the Type I transmitter and not greater than 1.0 amperes for the Type II transmitter.

3.2.1.1.2 Under-voltage protection. The transmitter shall not be damaged by external power voltages less than 24.0 Vdc, although specified performance is not required during the low-voltage periods. The transmitter shall automatically resume specified operation when the voltage returns to within the specified limits.

3.2.1.1.3 Warm-up time. Not greater than 5.0 seconds after application of primary power, the transmitter shall produce a detectable radio frequency (RF) output within the specified frequency range and stability limits. The transmitter shall achieve specified RF output power and all other performance requirements within 60 seconds.

3.2.1.1.4 Operating time. The transmitter shall meet all requirements of this specification for extended periods of time provided the transmitter base plate temperature is not greater than 71 degrees Celsius (°C).

3.2.1.1.5 Off-time between operating periods. The off-time between consecutive operating periods shall be not less than the time required for the transmitter base plate temperature to become not greater than 65°C.

3.2.1.1.6 Frequency range. The transmitter shall operate as specified on any selected crystal controlled frequency of not less than 225 Megahertz (MHz) and not greater than 260 MHz for Type I transmitters, and not less than 1435 MHz and not greater than 1540 MHz for Type II transmitters, by replacement of the crystal and adjustment of tuning devices. No other component replacement or adjustment shall be required to effect a frequency change within the range specified herein. The specific transmitter frequency setting shall be designated by the procuring activity or contracting officer (see 6.2.1).

3.2.1.1.7 Frequency stability. The stability of the transmitter RF carrier frequency shall be within ±0.003 percent of the assigned frequency.

3.2.1.1.8 RF output power. The transmitter RF output power shall be not less than 10.0 watts for Type I transmitters and not less than 2.0 watts for Type II transmitters, into a 50 ohm load with a voltage standing wave ratio (VSWR) of 2:1 ±10 percent varied through all phase angles.

3.2.1.1.9 Antenna system compatibility. The transmitter shall operate over the frequency range and specified RF output at any phase angle with an antenna system having a VSWR not greater than 3:1. The transmitter shall not be damaged nor its operation impaired, except for reduced RF output and increased antenna conducted spurious outputs, when operating into mismatched



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loads with a VSWR not greater than 5:1 over the frequency range and at any phase angle. The transmitter shall not be damaged by operating into a mismatched load with an infinite VSWR.

3.2.1.2 Modulation characteristics.

3.2.1.2.1 Modulation compatibility. The transmitter shall be capable of operating within the requirements of this specification when modulated by frequency modulation/frequency modulation (FM/FM) signals as described in IRIG Document 106-80, provided that the modulation frequency response of the transmitter is not exceeded.

3.2.1.2.2 Modulation input impedance. The input impedance of the modulator portion of the transmitter shall be not less than 50 Kilohms (K ohms) shunted by 30 picofarads (pf) maximum.

3.2.1.2.3 Modulation coupling. The modulation input of the transmitter shall be alternating current coupled.

3.2.1.2.4 Modulation response. The frequency response characteristic of peak-to-peak output deviation versus modulation input frequency shall be flat within ±2.0 decibels (dB) for modulation frequencies from 400 Hertz (Hz) to 250 kilohertz (KHz) for Type I transmitters and 400 Hz to 500 KHz for Type II transmitters, from a 10 KHz reference.

3.2.1.3 Carrier deviation.

3.2.1.3.1 Modulation sensitivity. The modulation sensitivity shall be 2.0 volts root mean square (Vrms) for 100 KHz deviation ±10 percent measured at 10 KHz.

3.2.1.3.2 Modulation linearity. The characteristic curve of peak-to-peak output frequency deviation versus input modulating voltage shall be linear within 2.0 percent of bandwidth for ±100 KHz deviation. Linearity shall be determined as the deviation from the best straight line by the least squares method.

3.2.1.3.3 Incidental frequency modulation. The peak-to-peak deviation of the unmodulated carrier due to incidental frequency modulation, crosstalk, or noise shall be not greater than 2 KHz for Type I transmitters and not greater than 5 KHz for Type II transmitters.

3.2.1.3.4 Spurious amplitude modulation. The peak-to-peak amplitude modulation of the carrier output signal shall be less than 5 percent of the carrier rms amplitude value when modulated to 100 KHz deviation.



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3.2.1.3.5 Spurious and harmonic outputs. All spurious and harmonic outputs measured in the transmission line between the transmitter output and the load shall be not less than 65 dB down in power from the carrier level to within 500 KHz of the carrier, and shall be not less than 40 dB below the carrier level from 500 KHz to 320 KHz of the carrier. This requirement shall apply to spurious components falling within the modulation bandwidth of the transmitter, as well as those harmonic and spurious components outside the modulation bandwidth of the transmitter from 0.150 MHz to 10,000 MHz. Signals conducted by power leads or radiated directly from units and cables shall be within the limits specified in MIL-STD-461, tests CE01, CE03, CE07, CS01, CS02, CS06, RE02, RS02, and RS03 (see 3.4.3).

3.2.2 Physical characteristics. The transmitter shall meet the physical requirements of Drawing 1356AS313.

3.2.3 Reliability. The reliability of the transmitter shall be not less than 2000 hours mean-time-between-failures (MTBF).

3.2.4 Maintainability. The maintainability program, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in MIL-STD-2078.

3.2.5 Environmental conditions.

3.2.5.1 Operational. The transmitter shall meet all requirements of 3.2.1 and 3.2.3 prior to, during, and after exposure to the following environmental conditions, separately or any combination thereof. The transmitter shall not exhibit any degradation in excess of limits specified herein as the result of exposure to any of the environments in a through f of Table I.

3.2.5.2 Non-operational. The transmitter shall meet all requirements of 3.2.1 and 3.2.3 after exposure to the environmental conditions c, g, and h of Table I, separately or any combination thereof. The transmitter shall not exhibit any temporary or permanent degradation in excess of limits specified herein as the result of exposure to any of the environments specified above.

3.2.6 Storage life. The storage life of the transmitter shall be such that it can be stored for 3 years in the shipping or storage container and then meet the requirements specified herein.

3.3 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3.1 Visual examinations. Test samples shall be subjected to visual examination to verify conformance to the physical and interchangeability requirements of this specification and the contract (see 4.6.1).



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TABLE I. Environmental conditions.

| PARAMETER | OPERATIONAL | NON-OPERATIONAL |
|---|---|---|
| a. Pressure | 70,000 ft (equivalent pressure) | ----- |
| b. High Temperature | +71°C | ----- |
| c. Low Temperature | -40°C | -40°C |
| d. Acceleration | 40 g, 3 axes | ----- |
| e. Random Vibration | 20-2000 Hz, .2$g^2$/Hz | ----- |
| f. Shock | 65g, 5ms, 1/2 sine | ----- |
| g. Thermal Shock | ----- | -55°C to +85°C |
| h. Moisture Resistance | ----- | 95% relative humidity |

3.4 Design and construction. Design and construction shall be in accordance with Drawing 1356AS313 and requirements specified herein (see 3.4.1).

3.4.1 Materials, processes, and parts. Fulfillment of major design objectives for the intended application specified herein shall be the prime consideration in the selection of materials, processes, and parts. In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other materials, processes, and parts requirements shall conform to MIL-E-5400.

c. Parts shall be selected using the following order of preference.

(1) Parts listed on a Qualified Parts List (QPL) with two or more sources.

(2) Parts listed on a QPL with only one source.



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(3) Commercial parts with two or more sources.

(4) Commercial parts with only one source.

3.4.1.1 Standard parts. Parts specified in 3.4.1c (1) and (2) are standard parts and do not require approval from the procuring activity. Parts selected for use from 3.4.1c (3) and (4) are considered nonstandard and require approval. Nonstandard parts approval requests shall be submitted on DD Form 2052 in accordance with the contract (see 6.2.2).

3.4.1.2 Commercial parts. Commercial parts specified in 3.4.1c (3) and (4) shall be readily available, nationally catalogued, off-the-shelf items.

3.4.1.3 Nonstandard parts. Use of nonstandard parts in first article or production units prior to government approval is at the sole risk of the contractor and subject to replacement if the government does not approve the proposed nonstandard part.

3.4.2 Electromagnetic interference. The transmitter shall meet the electromagnetic interference (EMI) requirements of MIL-STD-411, tests CE03, CE07, CE01, CS01, CS02, CS06, RE02, RS02, and RS03.

3.4.3 Nameplate and marking. The nameplate shall be as specified in Drawing 1356AS313. Interchangeable parts and assemblies shall be legibly marked. All markings shall be as specified in MIL-STD-130.

3.4.4 Serial numbers. Serial number assignments shall be as specified in MIL-N-18307.

3.4.5 Interchangeability. All assembly and subassembly level items of hardware that are subject to replacement and have the same part number shall be electrically, physically, and functionally interchangeable. Electrical component parts shall conform to the requirements specified in MIL-STD-965. The transmitter shall meet the interchangeability requirements of MIL-STD-454, Requirement 7, and the electrical and mechanical interface requirements of the AN/DKT-59 Telemetric Data Transmitting Set (MIL-T-05650).

3.4.6 Safety. The transmitter shall meet the safety requirements of MIL-STD-454, Requirement 1.

3.4.7 Human engineering. Crystals and tuning devices utilized to establish or change the RF carrier frequency shall be clearly identified. Replacement or adjustment of these items shall not be impeded by their position of other components or structures. Special equipment, tools or fixtures shall not be required to effect such frequency changes. All other transmitter controls and adjustments shall be set during production and initial testing and shall be inaccessible in the field.



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3.5 Soldering. Soldering practices shall be in accordance with MIL-STD-454, Requirement 5.

3.6 Workmanship. Fabrication and workmanship practices shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to monitor any of the inspections or to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall assure product conformance to the requirements, inspections and tests specified herein. The contractor's Quality Assurance Program shall be planned and conducted in a manner to achieve levels of quality and reliability commensurate with the intended application specified herein.

4.1.1.1 Inspection. The contractor's inspection system shall conform to MIL-STD-883 for both the commercial off-the-shelf microelectronics and new contractor-designed (peculiar) microelectronics.

4.1.1.2 Test equipment and inspection facilities. The manufacturer shall ensure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.1.2 Test plans and procedures. Detailed test plans and procedures in accordance with MIL-T-18303 shall be prepared by the contractor (see 6.2.2). The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary (see 6.2.1).



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4.1.3 Test data. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare test data showing quantitative results for all first article and quality conformance inspections (see 6.2.2) for each transmitter offered for acceptance. This report shall also identify all rejections from tests.

4.1.3.1 Electromagnetic interference test data. The contractor shall prepare a electromagnetic interference test report in accordance with MIL-STD-832 (see 6.2.2).

4.1.4 Failure reports. All test failures shall be documented in a failure report, identifying the cause of failure and the proposed corrective action for each failure (see 6.2.2).

4.1.5 Frequency change procedure. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare a frequency change procedure for replacement of the crystal and adjustment of the tuning devices at the Intermediate Maintenance Activity (IMA) level (see 3.2.1.1.6 and 6.2.2).

4.1.6 Reliability Reports. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare reliability test data, including failure reports, in accordance with MIL-STD-781 (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

(1) Visual examination (see 4.6.1)
(2) Performance test (see 4.6.2)
(3) Environmental test (see 4.6.3)
(4) Reliability test (see 4.6.4)

b. Quality conformance inspection (see 4.5)

(1) Individual test (see 4.5.2)
(2) Periodic production test (see 4.5.3)

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 4.3.3.

4.3.1 Maintenance and calibration. All test equipment shall be calibrated and maintained in accordance with MIL-STD-45662. Certification to this effect shall accompany all reports of first article and quality conformance tests (see 6.2.2).



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TABLE II. Individual tests.

| Examination or Test | Requirement Paragraph | Method Paragraph | | |
|---|---|---|---|---|
| | | First Article | Individual | Periodic Production |
| Visual examination | 3.3.1 | 4.6.1 | 4.6.1 | 4.6.1 |
| Performance tests | | | | |
| RF output open and short circuit protection | 3.2.1.1.1<br>3.2.1.1.7<br>3.2.1.1.8<br>3.2.1.1.9 | 4.6.2.1.1 | 4.6.2.1.1 | 4.6.2.1.1 |
| Warm-up and operational stability | 3.2.1.1.1<br>3.2.1.1.3<br>3.2.1.1.7<br>3.2.1.1.8<br>3.2.1.3.3 | 4.6.2.1.2 | 4.6.2.1.2 | 4.6.2.1.2 |
| Load mismatch | 3.2.1.1.1<br>3.2.1.1.3<br>3.2.1.1.7<br>3.2.1.1.8<br>3.2.1.3.5 | 4.6.2.1.3 | | |
| Low voltage operation | 3.2.1.1.7<br>3.2.1.1.8<br>3.2.1.3.1 | 4.6.2.1.4 | | |
| Modulation compatibility | 3.2.1.2.1 | 4.6.2.2.1 | | |
| Modulation input impedance | 3.2.1.2.2<br>3.2.1.2.3 | 4.6.2.2.2 | | |
| Modulation response | 3.2.1.2.4 | 4.6.2.2.3 | | 4.6.2.2.3 |
| Modulation sensitivity | 3.2.1.3.1 | 4.6.2.3.1 | 4.6.2.3.1 | 4.6.2.3.1 |
| Modulation linearity | 3.2.1.3.1<br>3.2.1.3.2 | 4.6.2.3.2 | 4.6.2.3.2 | 4.6.2.3.2 |
| Incidental frequency modulation | 3.2.1.3.3 | 4.6.2.3.3 | 4.6.2.3.3 | 4.6.2.3.3 |



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TABLE II. Individual tests (Continued)

| Examination or Test | Requirement Paragraph | Method Paragraph | | |
|---|---|---|---|---|
| | | First Article | Individual | Periodic Production |
| Spurious amplitude modulation | 3.2.1.3.4 | 4.6.2.3.4 | | 4.6.2.3.4 |
| Spurious and harmonic output | 3.2.1.3.5 | 4.6.2.3.5 | 4.6.2.3.5 | 4.6.2.3.5 |
| External power voltage variation | | | | |
| Steady-state voltage test | 3.2.1.1.1<br>3.2.1.1.2<br>3.2.1.1.3<br>3.2.1.1.7<br>3.2.1.1.8<br>3.2.1.3.1 | 4.6.2.3.6a | 4.6.2.3.6a | 4.6.2.3.6a |
| Transient voltage test | 3.2.1.1.7<br>3.2.1.1.8 | 4.6.2.3.6b | 4.6.2.3.6b | 4.6.2.3.6b |
| External power ripple test | 3.2.1.1.1<br>3.2.1.1.3<br>3.2.1.1.7<br>3.2.1.1.8<br>3.2.1.3.3 | 4.6.2.3.6c | | 4.6.2.3.6c |
| Frequency range | 3.2.1.1.6<br>3.2.1.1.7<br>3.2.1.1.8<br>3.2.1.2.4<br>3.2.1.3.2<br>3.2.1.3.5 | 4.6.2.3.7 | | 4.6.2.3.7 |
| Conducted and radiated emission | 3.2.1.2.1<br>3.2.1.3.5 | 4.6.2.3.8 | | |
| Interchangeability | 3.4.5 | 4.6.2.4 | 4.6.2.4 | 4.6.2.4 |
| Environmental test | 3.2.5 | 4.6.3 | | 4.6.3 |
| Reliability test | 3.2.3 | 4.6.4 | | 4.6.4 |



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4.3.2 Test equipment accuracy. Unless otherwise specified herein, all test equipment used shall have an accuracy within 20 percent of the tolerance for the variable to be measured. In some cases where an open-ended minimum or maximum requirement is given, the test equipment used shall be accurate to within five percent of that requirement.

4.3.3 Test environment. Unless otherwise specified in the applicable test procedure, all tests and inspections shall be conducted under the following environmental conditions:

a. Temperature (ambient): +23 ±10°C

b. Altitude: elevation of test facility

c. Relative humidity: 50 ±30 percent

4.3.4 Tolerances on environmental test conditions. The allowable tolerances on environmental test conditions shall be in accordance with MIL-T-5422.

4.4 First article inspections. When specified in the contract (see 6.2.1), first article inspections shall be conducted by the contractor on a transmitter representative of the production transmitters to be supplied under the contract. The first article sample (see 3.3) shall be subjected to the examinations and tests of Table II and shall be performed in the sequence specified therein.

4.4.1 Acceptance criteria. Failure of any transmitter to pass any first article test shall result in discontinuance of testing and rejection of the first article sample. The procuring activity reserves the right to require an additional sample and test if any transmitter fails to meet any requirement of this specification (see 6.2.1).

4.4.1.1 Post-tests. A transmitter which has previously failed first article inspection shall be repaired and retested in accordance with 4.6.2 prior to its resubmission for first article acceptance.

4.4.2 First article failure analysis and corrective action. The procuring activity shall be notified of a first article test failure and its probable cause (see 6.2.2).

4.5 Quality conformance inspections. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspections. All inspection and testing shall be subject to monitoring by the Government inspector. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance



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of the finished product. Quality conformance inspections shall consist of the following:

a. Individual tests (see 4.5.2)

b. Periodic production tests (see 4.5.3)

4.5.1 Quality conformance acceptance criteria. Any transmitter that fails to pass any individual test shall be rejected.

4.5.2 Individual tests. The individual tests shall be conducted on each transmitter offered for acceptance and shall consist of the tests in Table II.

4.5.3 Periodic production tests. After the first 50 transmitters are produced, a periodic sample consisting of one unit shall be selected at random from the first 10 transmitters of every 50 produced thereafter. This sample shall be selected from transmitters which have previously passed the individual tests of 4.5.2. The sample shall be subjected to all the examinations and tests specified in Table II.

4.5.3.1 Periodic production sample rejection. If the periodic production sample fails to meet the requirement of any examination or test specified herein, the sample shall be rejected. Before any further production lots or transmitters may be submitted for acceptance, a detailed report shall be prepared (see 4.1.4 and 6.2.2) describing the rejection and action taken to prevent recurrence of the defect which caused the failure. Further production lots will not be considered for acceptance until the periodic production sample has been approved.

4.6 Examination and test methods. Details of methods and procedures shall be in accordance with the approved test plan (see 4.1.2).

4.6.1 Visual examination.

4.6.1.1 Physical characteristics. The transmitter shall be inspected visually for conformance to 3.2.2.

4.6.1.2 Materials, processes, and parts. The transmitter shall be examined visually for conformance to 3.4.1.

4.6.1.3 Nameplates and product marking. The transmitter shall be inspected for conformance to 3.4.3.

4.6.1.4 Soldering. The transmitter shall be examined for conformance to 3.5.



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4.6.1.5 Workmanship. The transmitter shall be examined for conformance to 3.6.

4.6.1.6 Interchangeability. The transmitter shall be examined to verify physical interchangeability conformance to 3.4.5.

4.6.2. Performance tests. Unless otherwise specified, performance tests shall be performed as specified herein. The test transmitter shall be operated in conformance with 3.2.1.1.4 and 3.2.1.1.5. When tests are conducted with no modulation applied, the modulation input shall be grounded, and the transmitter load shall be a mismatched load with a VSWR of 2:1 ±10 percent with respect to 50 ohms and worst case phase angle adjusted at ±25°C. External power shall be in accordance with 3.2.1.1.1. The performance specified in 4.6.2.1 shall be continuously monitored during the conduct of test methods 4.6.2.2 and 4.6.2.3

4.6.2.1 Operational tests.

4.6.2.1.1 RF output open and short circuit protection. This test shall be performed prior to other tests specified herein. The transmitter RF output initially shall be terminated with either a short circuit or an open circuit. The transmitter shall be operated with a short circuit for 30 minutes and with an open circuit for 30 minutes. The transmitter shall be preconditioned (non-operating) at a temperature of +25°C for not less than 30 minutes prior to each 30 minute test of RF output open and short circuit protection. Baseplate temperatures shall be not greater than 71°C. Following each short circuit and open circuit operating period, the transmitter shall be operated with the load specified in 4.6.2 and the following verified:

a. Power demand as specified in 3.2.1.1.1

b. Frequency stability as specified in 3.2.1.1.7

c. RF output power as specified in 3.2.1.1.8

d. Antenna system compatibility as specified in 3.2.1.1.9

4.6.2.1.2 Warm-up and operational stability. The transmitter shall be operated under the conditions specified in 4.6.2. After preconditioning, the transmitter shall be operated for a period of 30 minutes. Beginning at external power turn-on and during the test period, the following shall be verified:

a. Power demand as specified in 3.2.1.1.1

b. Warm-up time as specified in 3.2.1.1.3



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c. Frequency stability as specified in 3.2.1.1.7

d. Output power as specified in 3.2.1.1.8

e. Incidental frequency modulation as specified in 3.2.1.3.3

4.6.2.1.3 Load mismatch. The transmitter shall be connected to a mismatched load having a VSWR of 3:1 (with phase angle between the resistive and the reactive load components varied between Ø and 360 degrees). The modulation input shall be grounded and external power shall be adjusted to +28.0 Vdc and the following verified:

a. External power as specified in 3.2.1.1.1

b. Warm-up time as specified in 3.2.1.1.3

c. Frequency stability as specified in 3.2.1.1.7

d. RF output power as specified in 3.2.1.1.8

e. Spurious outputs as specified in 3.2.1.3.5

The transmitter shall be connected to a load having a VSWR of 5:1 ±10 percent with the phase angle varied between Ø and 360 degrees and the following verified:

RF output may be less than as in 3.2.1.1.8

4.6.2.1.4 Low voltage operation. The transmitter shall be operated under the conditions of 4.6.2 without modulation. The external power shall be reduced to +16 ±0.5 Vdc for not less than 4 minutes. The external power shall then be restored to within the limits as specified in 3.2.1.1.1 and the following verified:

a. Frequency stability as specified in 3.2.1.1.7

b. RF output power as specified in 3.2.1.1.8

c. Modulation sensitivity as specified in 3.2.1.3.1

4.6.2.2 Modulation characteristics.

4.6.2.2.1 Modulation compatibility. The transmitter shall be tested for conformance with 3.2.1.2.1 by applying mixed IRIG subcarrier oscillators (VCOs) (channel bands 5 through 14 with a 3/2 power pre-emphasis slope) with rms level of 0.8 volts. The RF output shall be demodulated and examined to determine conformance with 3.2.1.2.1.



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4.6.2.2.2 Modulation input impedance. The transmitter shall be tested for conformance with 3.2.1.2.2 and 3.2.1.2.3.

4.6.2.2.3 Modulation response. The transmitter shall be supplied a swept (400 Hz to 250 KHz for Type I transmitters and 400 Hz to 500 KHz for Type II transmitters) audio signal of 2.0 Vrms and shall meet the requirement of 3.2.1.2.4.

4.6.2.3 Carrier deviation.

4.6.2.3.1 Modulation sensitivity. A test signal of 2.0 Vrms at 10 KHz shall be applied to the modulation input.

The modulation sensitivity shall be verified for conformance to 3.2.1.3.1.

4.6.2.3.2 Modulation linearity. Apply a test signal of 2.0 Vrms at 10 KHz frequency symmetrically about 0 volts. Linearity shall be determined by the least squares method by plotting curves of transmitter frequency versus modulation input for deviations from 400 Hz to 200 KHz for Type I transmitters and 400 Hz to 400 KHz for Type II transmitters, and the following verified:

a. Overall sensitivity as specified in 3.2.1.3.1

b. Modulation linearity as specified in 3.2.1.3.2

4.6.2.3.3 Incidental frequency modulation. The transmitter shall be operated under the conditions of 4.6.2 and the transmitter output shall be measured using a test demodulator with a 3 dB intermediate frequency (IF) bandwidth of not less than 300 KHz in conjunction with a 10 Hz-250 KHz ± 3.0 percent accuracy rms voltmeter. The voltmeter output shall be calibrated and recorded.

The incidental frequency modulation shall be verified for conformance to 3.2.1.3.3.

4.6.2.3.4 Spurious amplitude modulation. The transmitter shall be operated under the conditions of 4.6.2 except that the transmitter shall be modulated with a 10 KHz sine wave to produce a peak-to-peak carrier frequency deviation of 200 KHz.

The spurious amplitude modulation shall be verified for conformance to 3.2.1.3.4

4.6.2.3.5 Spurious and harmonic output. The transmitter shall be tested for spurious output by means of a spectrum analyzer capable of resolving 2 signals separated by not less than 10 KHz as follows:



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a. The product shall be operated into a mismatched load having a VSWR of 3:1 ±10 percent and with the phase angle varied between 0 and 360 degrees. The transmitter shall be modulated with a 12.5 KHz sine wave having an amplitude sufficient to produce a carrier frequency deviation of 100 KHz. The radiation spectrum from 10 KHz to 500 KHz away from the carrier shall be analyzed.

The spurious and harmonic outputs shall be verified for conformance to 3.2.1.3.5

b. The transmitter shall be operated into a 50 ohm attenuator connected to the spectrum analyzer. Radiation spectrums from 10 KHz to 500 KHz and from 0.150 MHz to 10,000 MHz shall be analyzed.

The spurious and harmonic outputs shall be verified for conformance to 3.2.1.3.5.

4.6.2.3.6 External power voltage variation. The external power voltage variation test shall consist of the following tests:

a. Steady-state voltage test: With no modulation input to the transmitter, external power shall be applied. The external power shall be adjusted to 24.0 +0.25,-0.0 Vdc and the following verified:

(1) Power demand as specified in 3.2.1.1.1

(2) Warm-up time as specified in 3.2.1.1.3

(3) Frequency stability as specified in 3.2.1.1.7

(4) RF output power as specified in 3.2.1.1.8

(5) Modulation sensitivity as specified in 3.2.1.3.1

With a primary power voltage of +32.0 +0.0,-0.25 Vdc applied, the test shall be repeated.

b. Transient voltage test: The transmitter shall be operated with an external power voltage of +28.0 ±0.5 Vdc for a period of 3 minutes while monitoring RF output power and center frequency. The external power voltage shall be reduced to 0 for not less than 1 second, raised to +40 ±1.0 Vdc for not less than 1 second, reduced to -40 ±1.0 Vdc



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for 1 second, then returned to +28.0 ±0.5 Vdc and the following verified:

(1) Frequency stability as in 3.2.1.1.7

(2) RF output power as in 3.2.1.1.8

c. External power ripple test: 28 Vdc external power having superimposed ripple with an amplitude of 3.0 Vrms and with a frequency spectrum from 50 to 15,000 Hz shall be applied to the transmitter and the following verified:

(1) Power demand as specified in 3.2.1.1.1

(2) Warm-up time as specified in 3.2.1.1.3

(3) Frequency stability as specified in 3.2.1.1.7

(4) RF output power as specified in 3.2.1.1.8

(5) Incidental frequency modulation as specified in 3.2.1.3.3

4.6.2.3.7 Frequency range. Utilizing the contractor's frequency change procedure (see 4.1.5), the transmitter frequency shall be changed from the contract or purchase order assigned frequency to 225 MHz and 260 MHz for Type I transmitters and 1435 MHz and 1540 MHz for Type II transmitters, during each test operation. The following shall be verified after each frequency change:

a. Frequency range as specified in 3.2.1.1.6

b. Frequency stability as specified in 3.2.1.1.7

c. RF output power as specified in 3.2.1.1.8

d. Modulation response as specified in 3.2.1.2.4

e. Modulation linearity as specified in 3.2.1.3.2

f. Spurious and harmonic outputs as specified in 3.2.1.3.5

4.6.2.3.8 Conducted and radiated emission. The transmitter sections shall be tested for conformance to 3.2.1.3.5 and 3.4.2.

4.6.2.4 Interchangeability. The transmitter shall be tested to verify electrical and operational interchangeability conformance to 3.4.5.



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4.6.3 Environmental tests. Environmental tests shall be performed as specified herein. When tests are conducted at high or low temperatures, test transmitters shall be stabilized at the specified temperature for not less than two hours prior to testing. If the transmitters are removed from the chamber for any portion of the tests, such tests shall be completed within 10 minutes of removal from the chamber. If the tests are not completed within the 10 minute period, the transmitters shall be returned to the temperature chamber for not less than two hours before continuing the tests. The test transmitters shall stabilize at room ambient conditions prior to performance of post tests and measurements.

4.6.3.1 Operational environmental tests. The transmitter performance shall be tested as specified in 4.6.3.1.1 through 4.6.3.1.6 and meet the requirements of 3.2.1.1.1 through 3.2.1.3.5 during and after exposure to the environmental conditions specified in Table I, a through f.

4.6.3.1.1 Reduced pressure. The transmitter shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.2 High temperature. The transmitter shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.3 Low temperature. The transmitter shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.4 Acceleration.

4.6.3.1.5 Random vibration. The transmitter shall be tested with a spectral power density of 0.2 $g^2$/Hz from 20 to 2000 Hz for two hours in each of three mutually perpendicular axes.

4.6.3.1.6 Shock. The transmitter shall be tested in accordance with MIL-T-5422, Shock Test.

4.6.3.2 Nonoperational environmental tests. The transmitter's performance shall be tested as specified in 4.6.3.2.1 through 4.6.3.2.3 and meet the requirements of 3.2 after exposure to the environmental conditions specified in Table I, c, g, and h.

4.6.3.2.1 Low temperature. The transmitter shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 equipment Class 2 at -40°C for 48 hours.



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4.6.3.2.2 Thermal shock. The transmitter shall be tested in accordance with MIL-T-5422, Temperature Shock Test.

4.6.3.2.3 Moisture resistance. The transmitter shall be tested in accordance with MIL-T-5422, Humidity Test.

4.6.4 Reliability test. The transmitter shall be tested as specified in the Probability Ratio Sequential Test (PRST) Plan VIIIC, Figure C-8, Appendix C of MIL-STD-781 to demonstrate conformance to 3.2.3.

4.6.4.1 Reliability acceptance criteria. The Government inspector shall be notified any time the total test hours and test failures plotted in accordance with the PRST Acceptance and Reject criteria for Test Plan VIIIC of MIL-STD-781 reflect a "reject" situation during the conduct of the contractor's approved production test procedures. The procuring activity reserves the right to stop the acceptance of products at any time a reject situation exists pending a review of the contractor's efforts to improve the product, product components and parts and the product workmanship.

4.7 Production transmitters. Transmitters supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article. Each transmitter shall be capable of successfully passing the same tests as imposed on the first article. Evidence of noncompliance with the above shall constitute cause for rejection, and, for transmitters already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections.

4.8 Packaging inspection. The preservation, packing and marking of the transmitter shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packaging, and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), the level of preservation-packaging, and packing shall be as specified in MIL-E-17555 and MIL-STD-794, Level C. The method of preservation for Level C shall be determined in accordance with the selection chart in Appendix D specified in MIL-STD-794 and the method of packaging shall be as specified in MIL-P-116.

5.2 Marking.

5.2.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).



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5.2.2 Normal marking. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3 Spare parts. When specified (see 6.2.1), spare parts shall be preserved and packaged to the same specifications as a complete transmitter product except that weight of spare parts will determine selection of style of the box or container.

6. NOTES

6.1 Intended use. The T-1458/DKT-59 radio transmitter is intended for use in the AN/DKT-59 Telemetry Data Transmitting Set.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Number, title, and date of this specification.

b. Number and type of transmitters to be procured.

c. First article inspection if different from 4.4.

d. Test plans and procedures if different from 4.1.2.

e. Facility assigned to perform inspections (see 4.1).

f. Test data and reports if different from 4.1.3.

g. Acceptance criteria if different from 4.4.1 and 4.5.1.

h. Inspection conditions if different from 4.3.

i. Test equipment accuracy if different from 4.3.2.

j. Level of packing and packaging if different from 5.1.

k. Special marking (see 5.2.1).

l. Spare parts preservation and packaging (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered



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in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

| Paragraph number | Data requirements | Applicable DID number | Options |
|---|---|---|---|
| 4.1.2 | Procedures, Test | UDI-T-21347 | ---- |
| 4.1.3 | Reports, Test | DI-T-2072 | ---- |
| 4.1.3.1 | Electromagnetic Interference Test Report | DIR-7062 | ---- |
| 4.1.4 | Report, Failed Item Analysis | DI-R-7039 | ---- |
| 4.1.5 | Manual, Technical, Standard | DI-M-2044 | ---- |
| 4.1.6 | Reports, Reliability Test and Demonstration | DI-R-7034 | ---- |
| 4.3.1 | Equipment Calibration Procedures | DI-R-7065 | ---- |
| 4.4.2 | Failure Analysis and Corrective Action Report (First Article) | DI-R-5299 | ---- |

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)



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6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, and test approval of the first article sample.

Preparing activity:
Navy - AS
(Project No. 5815-N118)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions - Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-T-85653(AS) | Transmitter, Radio T-1458/DKT-59 |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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